

Fluoroquinolone Resistance Mutations among *Mycobacterium tuberculosis* and their Interconnection with Treatment Outcome

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Abstract

Background: Fluoroquinolone (FQ) antibiotics are among the most potent second-line antitubercular drugs these days. The aim of the study was to analyze the frequency and pattern of genetic mutation in preextensive (pre-XDR) and extensively drug-resistant *Mycobacterium tuberculosis* using second-line line probe assay (LPA) and to compare drug-resistant mutations with different treatment outcomes. **Methods:** Sputum, lymph node aspirate, and cold accesses from patients with rifampicin-resistant Tuberculosis (TB) were subjected to first-line and second-line LPA (Genotype MTBDRsl by Hain Life Science, Germany) to assess additional drug resistance to fluoroquinolones (levofloxacin and moxifloxacin). Final treatment outcomes as per the National TB Elimination Program were assessed and compared with the mutation profile. **Results:** One hundred and fifty subjects were observed to have mutations associated with resistance to FQs and constituted the final study population. The most frequent mutation observed among GyrA drug resistance mutation was D94G (Gyr A MUT3C, 44/150, 66%) corresponding to high-level resistance to levofloxacin and moxifloxacin. The same mutation was associated with poor treatment outcome as died or treatment failure (odds ratio 2.50, relative risk 1.67, $P = 0.043$). The most common hetero-resistance mutation pattern observed in GyrA gene was wild type plus Asp94Gly mutation in 24.6% of isolates. **Conclusions:** GyrA MUT3C hybridization corresponding to single-point mutation of aspartic acid to glycine at codon 94 constitutes the most common mutation in GyrA gene locus in *M. tuberculosis* with significant association with treatment outcome as died compared to those with treatment outcome as cured.

Keywords: Fluoroquinolones, gene mutation, treatment outcome

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INTRODUCTION

Tuberculosis (TB) is a communicable disease with significant morbidity and mortality worldwide. Until the coronavirus (COVID-19) pandemic, TB was the leading cause of death from a single infectious agent, ranking above HIV/AIDS.^[1] Globally, the estimated number of deaths from TB increased between 2019 and 2021, reversing years of decline between 2005 and 2019.^[1] In 2021, there were an estimated 1.4 million deaths among HIV-negative people and 187,000 deaths among HIV-positive people. The estimated incidence of multidrug-resistant (MDR)/rifampicin-resistant tuberculosis (RR-TB) (as per the Global TB Report 2022) in 2021 for India was 119,000 (93,000–145,000). The year 2022 saw an increase of 32% in the number of MDR/RR-TB cases detected under the National Tuberculosis Elimination Program (NTEP) as compared to 2021.^[2] Globally also the number of people developing MDR/RR-TB increased from

437,000 per year in 2020 to 437,000 per year in 2021 (World Health Organization [WHO] Global TB Report 2022).

Rapid and accurate detection of MDR/XDR-TB is of important as this allows for optimizing therapy and reducing transmission. This is best achieved using molecular tests such as Cartridge-Based Nucleic Acid Amplification Assay and Line probe assay. MTBDRsl version 2.0 assay is one such line probe assay (LPA) done for second-line resistance detection

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in MDR TB/RR-TB in India under programmatic conditions. One study found that among fluoroquinolone (FQ)-resistant isolates, 18 different types of banding patterns corresponding to mutations were observed.^[3] The most common banding pattern was MUT3C followed by GyrA MUT1 and GyrA MUT3A, respectively.^[3]

Studies relating to different mutations to the treatment outcome in pre-XDR-TB patients are very limited. The present analysis is an attempt to identify the frequency and pattern of these mutations associated with FQ resistance and to evaluate different outcomes related to such mutations.

METHODS

This retrospective analysis was carried out at the Nodal drug-resistant tuberculosis center at the Department of Respiratory Medicine and Intermediate Reference Laboratory. Intermediate Reference Laboratory (IRL) laboratory is certified by Central TB Division, India, for the testing of molecular and phenotypic drug sensitivity of *Mycobacterium tuberculosis*.

The Genotype MTBDRsl version 2.0 assay was performed as per the manufacturer's instructions. In brief, 1 mL of positive liquid culture was centrifuged and the pellet was taken for DNA extraction by GenoLyse kit Hain Lifescience (Nehren, Germany). Before amplification, the kit components AM-A and AM-B were mixed, and then, the extracted DNA was added and amplified. The hybridization was performed using TwinCubator/GT-Blot and the results were analyzed. The strip contains 27 probes to check internal controls, identification of *M. tuberculosis* complex, and drug targets GyrA, GyrB for FQ and *rrs*, *eis* for second-line injectable drug. The missing wild probe and the presence of mutant probe were considered as resistant.

A total of 250 consecutive drug-resistant isolates (either MDR TB/RR TB or isoniazid resistant-TB) received between 2020 and 2022 were analyzed by routine LPA testing at IRL. Upon completing second-line LPAs (SLLPAs) of these isolates, FQ resistance was detected in 150 (60%) and mutations in these isolates are studied and compared with the final treatment outcome.

A phenotypic drug sensitivity test (pDST) to moxifloxacin was done in all patients with FQ resistance on SLLPA. 106/150 isolates (70.6%) were resistant to moxifloxacin on pDST. 37/150 FQ resistant isolates on SLLPA were sensitive on pDST and 7 samples showed no growth on culture.

Treatment regimen – 132/150 patients with FQ resistance were put on all longer oral M/XDR-TB regimens containing bedaquiline (BDQ)/moxifloxacin/linezolid (LZD)/clofazamine (CFZ)/Cycloserine (CS). 10/150 patients, who were rifampicin sensitive and H + FQ resistant were initiated on moxifloxacin/R/Z/E. Eight isolates were resistant to MUT 3B/3C/3D and so were put on a combination regimen of BDQ/Delamanid/CFZ/CS/LZD.

Treatment outcomes were assigned as per NTEP guidelines. The data analysis was performed using Epi Info version 7.2.1.0 (CDC, Atlanta, Georgia, US). Statistical significance was established at $P = 0.05$. The study was started after obtaining institutional ethical committee approval.

RESULTS

Two hundred and fifty rifampicin-resistant samples were subjected to SLLPA and FQ resistance was detected among 150 (60%). In the study group, 67% (101/150) were male patients and 33% (49/150) were female patients. Out of total 150 subjects, 130 (86.6%) were from rural area where as 20 (13.3%) were from urban area. Seventy-eight percentage (116/150) of the subjects were underweight with body mass index <18.5 and 20% (31/150) were in the normal range and 2% (3/150) were overweight. Forty-five percentage (67/150) of subjects were severely underweight. One hundred and forty-three were sputum samples, four were lymph node needle aspirate two cold abscesses and one was psoas abscess aspirate. In this study, all subjects were subjected to blood sugar and HIV antibodies estimation. Four subjects were reactive for HIV antibodies with normal CD4 count and 3 were diabetic controlled on drugs.

Detection of fluoroquinolone resistance

The frequency and mutation profiling of FQ resistance were determined at Gyr gene. In total, 149 out of 150 showed resistance for GyrA gene while GyrB gene mutations was present in only one case with a mutation at GyrB WT1. Figure 1 shows all types of mutations in the GyrA gene and the pattern of corresponding wild and mutant bands. The absence of GyrA WT3 band (56%, 84/150) and the development of MUT3C (44%, 66/150) was the most common pattern of mutation. It was followed by GyrA MUT1 (18.7%, 28/150) and GyrA MUT3B (18%, 27/150). Out of 150, 94 samples showed heteroresistance, i.e. both wild-type and mutant alleles were detected. The most common heteroresistance mutation pattern was a combination of wild type and Asp94Gly (WT plus D94G) in 37 (24.6%) isolates. This was followed by wild type and Ala90Val (WT + A90V) in 18 (12%) isolates and wild type and Asp94Asn/Tyr in 15 (10%) isolates (WT + D94N/Y). WT plus Asp94His (WT + D94H) was the least common hetero-resistance pattern observed in this study (2%).

Fluoroquinolone resistance mutations and treatment outcome

Table 1 shows the treatment outcomes of the study subjects. Among 150 subjects, 61 were still on treatment, so the outcome could be analyzed among 89 patients only. Among these 89 patients, 49 (5%) patients had an outcome as cured and 28 (31.4%) subjects had an outcome as died, 6 (6.7%) had an outcome as treatment failure and 7 (7.8%) were lost to follow-up. The most common mutant pattern observed was GyrA MUT3C corresponding to D94G mutation in 46.9% of cured patients. Only one wild type mutation in GyrB gene locus was observed in cured among 150

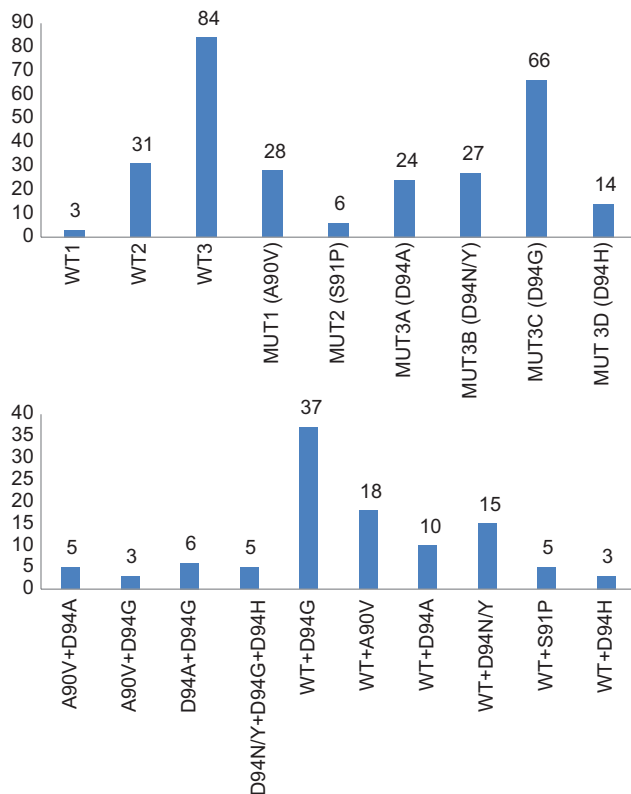


Figure 1: Bar diagram showing the frequency of GyrA mutations ($n = 150$)

isolates analyzed in this study. The most frequent mutation observed among 28 died subjects was GyrA MUT3C seen in 13 isolates (46.4%). The most frequent mutation pattern observed among treatment failure was that of GyrA MUT3C in 4 (66%) patients.

Heteroresistance in fluoroquinolone and treatment outcome

Table 2 shows a comparison of GyrA heteroresistance mutations with treatment outcome. Two patients had both MUT1 and MUT3C and both had a poor outcome but statistical significance could not be established ($P = 0.09$).

The presence of any GyrA Mutation was associated with poor treatment outcomes (died and treatment failure) as compared to those without the same mutation (odds ratio [OR]: 3.71 [confidence interval (CI): 1.86–7.42] $P = 0.0002$, Table 3). The OR for patients with outcome as died or treatment failure among subjects with hetero-resistance mutation was calculated to be 0.659 (0.402–1.079) with $P = 0.106$. On comparison of GyrA MUT3C mutations with treatment outcome, patients having D94G mutation (GyrA MUT3C) were observed to have 1.67 times risk of the worse treatment outcome (death or treatment failure) as compared to patients without mutation (OR - 2.50, CI = 1.026–6.088). GyrA D94G mutation (MUT3C) was significantly associated with treatment outcome as died or treatment failure (53%) as compared to patients with outcome as cured (46.9%) ($P = 0.043$).

Table 1: Comparison of outcome among fluoroquinolone-resistant isolates ($n=89$)

Mutation	Cured ($n=49$), n (%)	Died ($n=28$), n (%)	Failure ($n=6$), n (%)	P
GyrA WT1	2 (4.1)	0	0	0.481
GyrA WT2	16 (32.6)	4 (14.0)	1 (16.7)	0.158
GyrA WT3	29 (59.1)	17 (60.7)	4 (66.7)	0.933
GyrA A90V	11 (22.4)	5 (17.8)	1 (16.7)	0.836
GyrA S91P	3 (6.1)	1 (3.5)	0	0.737
GyrA D94A	4 (8.1)	3 (10.7)	1 (16.7)	0.785
GyrA D94N/Y	9 (18.3)	6 (21.4)	1 (16.7)	0.957
GyrA D94G	23 (46.9)	13 (46.4)	4 (66.7)	0.043
GyrA D94H	4 (8.1)	3 (10.7)	1 (16.7)	0.785
GyrB WT1	1 (2)	0	0	0.697
GyrB N538D	0	0	0	0

Table 2: Frequency of GyrA multiple mutations and treatment outcome

Mutation	Number of patients		P
	Cured	Died/treatment failure	
D94A + D94G	1	1	0.809
D94N/Y + D94H	3	3	0.667
D94N/Y + D94G + D94H	3	3	0.667
D94N/Y + D94G	5	3	0.802
A90V + D94N/Y	1	0	0.395
A90V + D94G	0	2	0.09
A90V + D94H	1	0	0.395
WT + A90V	10	5	0.470
WT + S91P	3	1	0.488
WT + D94A	3	2	0.938
WT + D94N/Y	8	4	0.527
WT + D94G	18	8	0.175
WT + D94H	2	1	0.766
WT + D94N/Y + D94G	4	0	0.08

DISCUSSION

Drug resistance causes prolonged treatment periods and increases the risk of poorer treatment outcomes in TB.^[4] The role of the FQ is implied in how extensively drug-resistant (XDR) TB is defined, i.e. as TB caused by *M. tuberculosis* strains that fulfill the definition of MDR/RR-TB and are also resistant to any FQ (levofloxacin or moxifloxacin) and at least one additional Group A drug (either BDQ or LZD or both).^[5] FQs vary in their potency against *M. tuberculosis* with 3rd- and 4th-generation agents (e.g., levofloxacin, gatifloxacin, and moxifloxacin) having higher activity than 2nd-generation agents (e.g., ciprofloxacin or ofloxacin).^[6,7]

The global coverage of testing for resistance to FQs remains much lower, being 50% in 2021.^[8] The rate of FQ resistance prevalence among non-MDR-TB is higher than 0.8% of the global estimate, which may be a warning against the widespread use of FQ in the community.^[9] As per the WHO

Table 3: Comparison of GyrA mutation patterns with different treatment outcomes

	Mutation	Cured, <i>n</i> (%)	Died/failure, <i>n</i> (%)	OR for outcome died/failure (95% CI)
Any Gyr A	Present	48	34	3.71 (1.86–7.42)
	Absent	84	16	<i>P</i> =0.0002
Gyr A Wild + mutant (hetero resistance)	Present	35 (65)	19 (35)	0.659 (0.402–1.079)
	Absent	14 (47)	16 (53)	<i>P</i> =0.106
GyrA MUT3C (D94G)	Present	16 (33)	20 (55)	2.500 (1.026–6.088)
	Absent	32 (66)	16 (44)	<i>P</i> =0.043
Z statistic: 2.018				
RR: 1.67				

OR: Odds ratio, CI: Confidence interval, RR: Relative risk

report in 2018, 8.5% of all MDR-TB patients had additional resistance to at least one drug.^[10] In the present study, FQ resistance among MDR-TB estimates to be 60% (150/250) which is higher than global estimates.

Mutation at codon 94 can be identified by the only commercial molecular test currently available (Genotype MTBDRsl; Hain Lifesciences) by the absence of the WT3 probe, whereas 94Gly mutations can be detected specifically as MUT3C, and 94Ala as MUT3A. The frequency of GyrA gene mutation (99.3%) was found to be higher than that of GyrB (0.6%) in the present study which is consistent with the data reported by Kabir *et al.*^[11] from Pakistan. The finding that the development of MUT3C (44%, 66/150) was the most common pattern of mutation in FQ-resistant TB isolates is higher than the figure reported by Sethi *et al.*^[3] (33%) from Chandigarh, India. It was followed by GyrA MUT1 (18.7%, 28/150) and GyrA MUT3B (18%, 27/150) and the pattern was consistent with that reported by Kabir *et al.*^[11]

The most common heteroresistance mutation pattern in GyrA gene in the present study was a combination of wild type and Asp94Gly (WT plus MUT3C) in 37 (39.3%) of isolates. This was consistent with a study by Rigouts *et al.*^[12] in which the same pattern of hetero resistance was observed in 4 out of 10 (40%) heteroresistance patterns. The least common hetero resistance pattern observed was WT + D94H in 3 (3.1%) isolates. Among 37 isolates having WT + Asp94Gly mutation, 8 had a treatment outcome as died or failure whereas the rest 18 had an outcome as cured (*P* = 0.175). Subjects with a double mutation of Ala90Val and Asp94Gly (GyrA MUT1 + MUT3C) were all had an outcome of died/failure (*P* = 0.09). Heteroresistance has been reported to be responsible for limited treatment options and increasing rates of treatment failure in TB patients as reported by Rigouts *et al.*^[12] In our study, two patients had heteroresistance at both MUT1 and MUT3C and both had a poor outcome but statistical significance could not be established (*P* = 0.09).

The most frequent mutation in GyrA gene locus among died patients was MUT3C in 46% of patients (13/28). MUT3C was the most frequent mutation observed among treatment failure cases also (66%, 4/6). The present study compared treatment outcome of different mutations and statistically significant difference was noted in subjects with GyrA

MUT3C. Death or failure was significantly higher among those patients (*P* = 0.043). GyrA heteroresistance mutations were compared to treatment outcomes but no significant difference could be made [OR: 0.659 (CI = 0.402–1.079), *P* = 0.106].

This study observed the presence of GyrA Mutations to be associated with poor treatment outcome (died and treatment failure) as compared to those without the same mutation (OR: 3.71 [CI: 1.86–7.42] *P* = 0.0002). This was comparable to a study by Liu *et al.*^[13] from Eastern China who also observed GyrA gene mutation associated with poor outcome (*P* = 0.002). Statistical analysis of cured and bad treatment outcome (died or treatment failure) was also done and it was observed that the subjects with this mutation had 1.67 relative risk of outcome as died or failure than those without the mutation (OR: 2.500, CI = 1.026–6.088, *P* = 0.043). Rigouts *et al.*^[12] from France reported a higher OR 4.3 (CI 1.4–13.0) of Asp94Gly mutation predicting a significantly lower proportion of cure compared to the present study. The specific impact of GyrB mutation on patient outcome could not be analyzed because only one sample out of 150 isolates had GyrB mutation.

The development of mutation leading to resistance to a particular drug does not exclude it from the treatment regimen. The A90V (GyrA MUT 1), S91P (GyrA MUT2), and D94A (GyrA MUT3A) mutations led to resistance against levofloxacin but a higher generation of FQ (i.e., moxifloxacin) can be used at a higher dose. Both levofloxacin and moxifloxacin are ineffective if D94G mutation is present.

CONCLUSIONS

The most frequent fluoroquinolone resistant mutation was observed to be Gyr A mutation D94G (Gyr A MUT3C) corresponding to the high-level resistance to levofloxacin and moxifloxacin in this study from western part of India. Presence of any Gyr A mutation was observed to be associated with poor treatment outcomes (died and treatment failure) as compared to those without this mutation. Gyr A D94G mutation (MUT3C) was significantly associated with treatment outcome as died or treatment failure (53%) as compared to patients with outcome as cured. Among hetero-resistance mutation patterns, subjects with a double mutation of Ala90Val and Asp94Gly (Gyr A MUT1 +MUT3C) mostly had outcome as died/failure.

Strengths

The present study compares the FQ resistance mutation pattern among TB patients in Western India for the first time. The correlation of FQ-resistant mutations with treatment outcomes had not been studied earlier. Heteroresistance among FQ was also analyzed and compared with treatment outcomes, making this study unique.

Limitations

The limitation of this study lies in the absence of sequencing data for confirmation of different drug-resistant-related mutations. LPA is less efficient than conventional culture-based drug sensitivity test in detecting resistance in samples that harbor both drug-susceptible and-resistant bacteria (i.e. heteroresistance). Moreover, there were less extra-pulmonary samples for mutation profile analysis and hence effect of mutations in extra-pulmonary TB could not be studied in detail. Impact of comorbid illnesses could not be analyzed as there were a few cases with such illnesses. This study focused only on the FQ mutations as data on additional drug resistance was unavailable and therefore their impact on treatment outcome could not be studied. Being a hospital-based retrospective study, findings cannot be generalized to the community at large, but despite the above limitations, it gives valuable information regarding the mutation trends of FQ-resistant TB cases and associated treatment outcomes.

Ethical statement

The study was approved by the Institutional Ethics committee of Jawarlal Nehru Medical College, Ajmer (Approval number 1166).

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Conflicts of interest

There are no conflicts of interest.

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